2D Finite Element Analysis of a Microstrip Line

Capacitance and impedance, with a measured error

Group C

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MTH321 Numerical PDEs — Project 2

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A number, and how wrong it is

What is required

- **Capacitance per unit length** of the cross-section.
- The **all-air capacitance** on the **same mesh**.
- $\varepsilon_{\mathrm{eff}}$ and Z_0 from that matched pair.

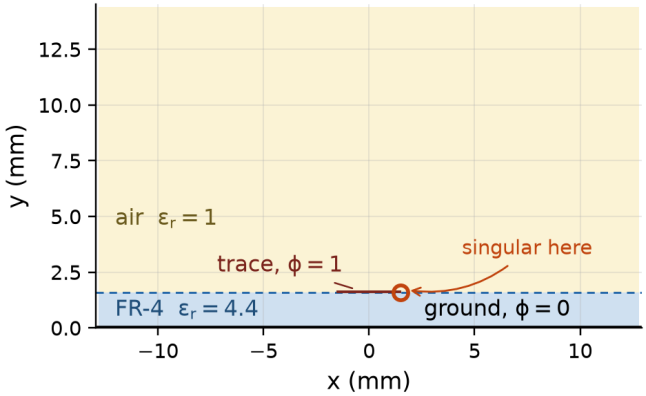
What would not be enough

- One FEM number that looks close to $50\,\Omega$.
- Agreement with an empirical formula only.
- Calling the result converged because two meshes agree.

Takeaway

The deliverable is a number together with the distance from its limit.

The cross-section and the equation



Electrostatic model

$$-\nabla \cdot (\varepsilon \nabla \phi) = 0$$

$\phi = 1\,\mathrm{V}$ on the trace, $\phi = 0$ on the ground

Zero normal flux on the artificial top and side walls.

Dimensions

$H = 1.6\,\mathrm{mm}$, $w = 3.0\,\mathrm{mm}$, $t = 0.035\,\mathrm{mm}$, substrate $\varepsilon_r = 4.4$.

Takeaway

The field is singular where the conductor edge ends on the dielectric interface.

Weak form and P_1 elements

Weak problem

Find $\phi \in V_g$ with $\int_\Omega \varepsilon \nabla\phi \cdot \nabla v \, d\Omega = 0$ for every $v \in \mathcal{V}$.

Discretisation

- Conforming triangles, **linear shape functions**.
- The trace volume is excluded; **Dirichlet nodes are eliminated**.
- **Positive** ε makes the free system SPD.
- Capacitance from the energy, $C = 2U/V_0^2$.

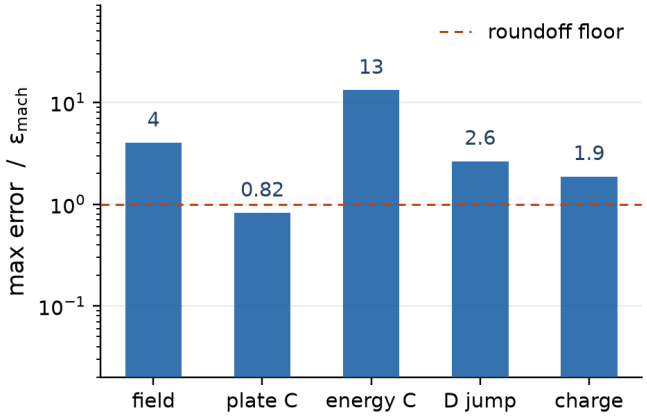
Interface

Potential is continuous and the normal flux balance holds in the **weak** sense only. Discrete fluxes need not match pointwise across an edge.

Takeaway

A small residual proves the linear system was solved, not that the mesh is adequate.

23 analytic checks, one verdict



What is checked

- Uniform field against $\phi = y/d$, for $\varepsilon_r = 1$ and 4.4.
- C against $\varepsilon_0 \varepsilon_r A/d$ and a two-layer series.
- Normal D continuous across the interface.
- Charge balance and zero flux on the far walls.
- Symmetry, definiteness, monotone energy under refinement.

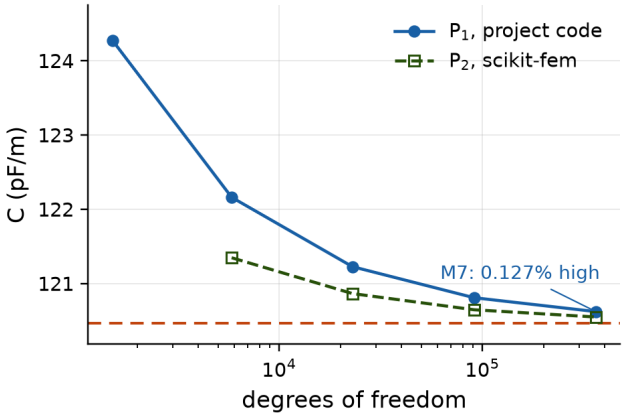
Result

All pass; the worst error is $\approx 13\,\varepsilon_{\mathrm{mach}}$.

Takeaway

Assembly and extraction are exact to roundoff, so every error left is discretisation.

Two elements, two codes, one limit



Two independent routes

- The project's P_1 sequence, **Richardson extrapolated**.
- An independent scikit-fem P_2 solve on the **same meshes**.

$$120.4673 \text{ vs } 120.4670 \ \mathrm{pF/m}$$

They agree to 2.9×10^{-6} relative.

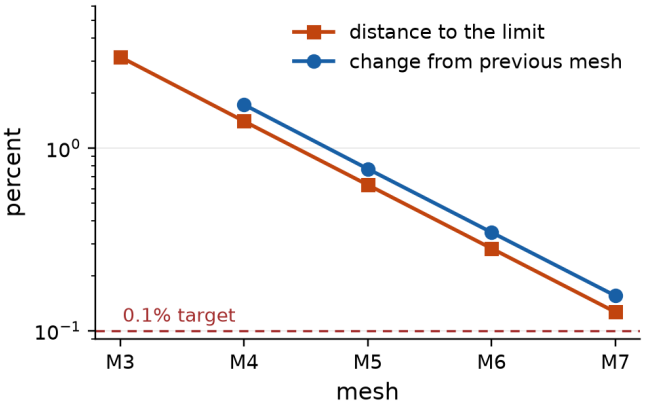
Reference adopted

$C_{\mathrm{ref}} = 120.467\,\mathrm{pF/m}$.

Takeaway

With a reference, an error can be measured instead of guessed.

The old metric measured the wrong thing



Two different questions

- Change from M6 to M7: 0.156%.
- Distance from the limit at M7: 0.127%.

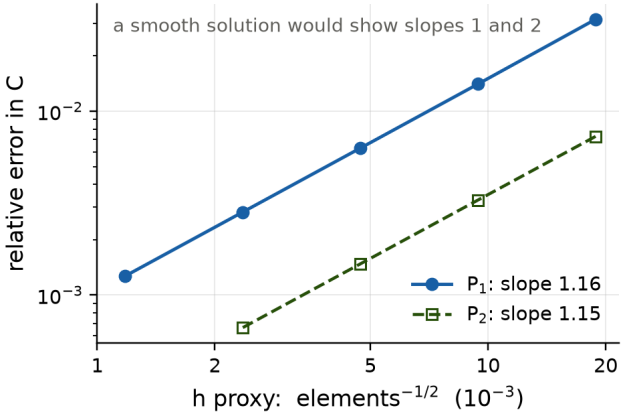
Why they differ

For order p the change is $(2^p - 1)$ times the error, here $2^{1.16} - 1 = 1.23$. A shrinking change bounds nothing without knowing p.

Takeaway

M7 is 0.127% from the limit, so the 0.1% target is missed — it is not converged.

Quadratic elements do not buy an order



Fitted slopes

Family	DOF range	slope
P_1	1.5k–362k	1.16
P_2	5.8k–362k	1.15

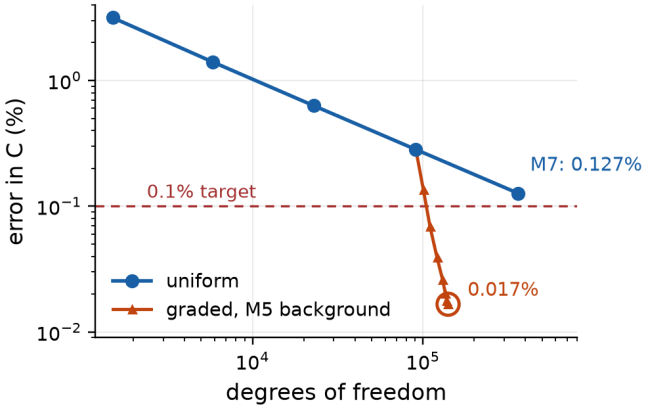
Diagnosis

Doubling the polynomial degree leaves the rate unchanged, so the limit is not the basis. The fitted field exponent near the corner is 0.54, giving an energy error of order $2 \times 0.54 \approx 1.1$.

Takeaway

The order is set by the corner, so uniform refinement is the wrong tool.

Corner grading restores the order



Same model, graded partition

Geometric refinement around the two trace ends on fixed background meshes M2 to M5.

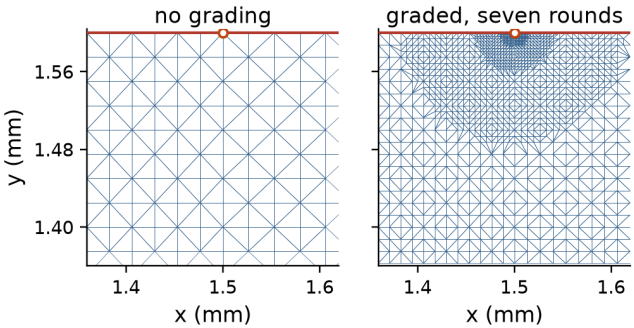
Result on the M5 background

	value
DOF	139,426
error	0.017%
vs M7	7.6× better
cost	39% of M7

Takeaway

The plateau falls fourfold per background level, so the remaining error is h^2 again.

Where the extra elements go



The same M5 background, seen near the right trace end.

What changed

- Elements are marked by **distance to the corner** and refined.
- The far field and the middle of the trace keep their coarse mesh.
- The operator, the boundary conditions and $C = 2U/V_0^2$ are identical.

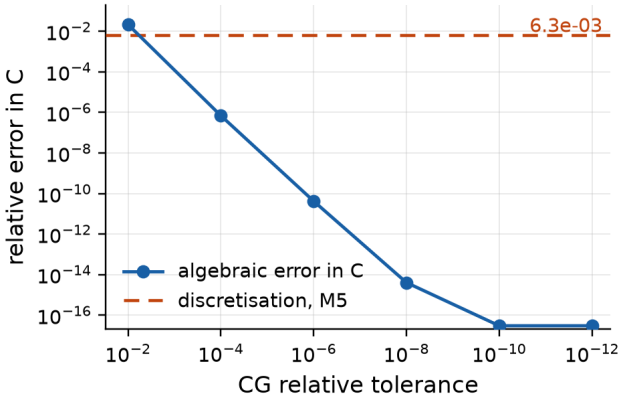
Cost

The solve stays a sparse direct solve; only the unknowns near the corner grow.

Takeaway

Grading changes the partition, not the equations — so it is a choice we may make.

The linear solver is not the bottleneck



Tolerance ladder on M5

- Conjugate gradients from 10^{-2} down to 10^{-12}.
- The capacitance error in C falls with the tolerance until 10^{-10} and then stays at roundoff.

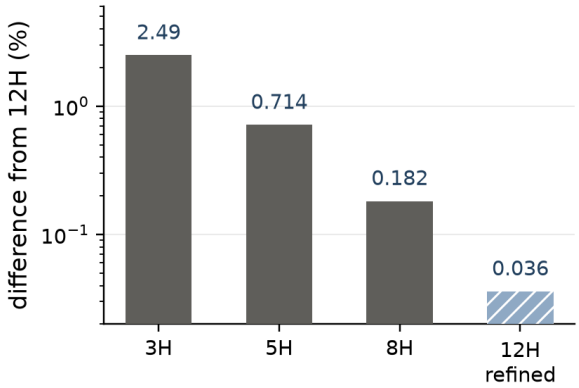
Two errors, far apart

Discretisation error on the **same mesh**: 6.3×10^{-3}.
Algebraic error at 10^{-10}: below 10^{-16}.

Takeaway

Thirteen orders of magnitude separate them: the budget belongs to the corner.

The $8H$ box costs 0.18%



Truncation against the 12H box

	3H	5H	8H
difference	2.49%	0.71%	0.18%

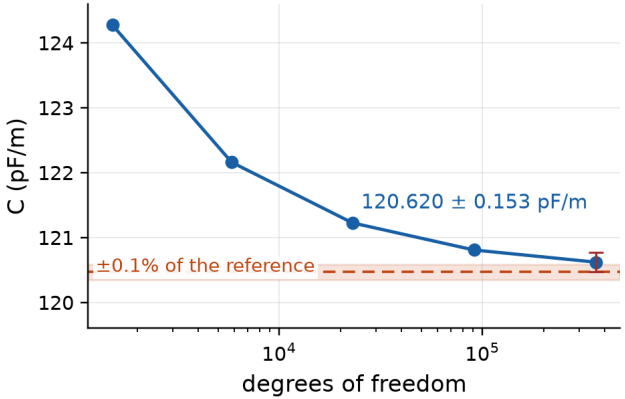
Mesh adequacy, separately

Refining the 12H mesh moves it by 0.036% only, so the 0.18% is truncation and not mesh noise.

Takeaway

Truncation (0.18%) and discretisation (0.127%) are the same size: quote both.

The capacitance, with its error bar



Submitted values

$$C = 120.620 \pm 0.153 \text{ pF/m}$$

$C_{\text{air}} = 35.968 \text{ pF/m}$ (same M5 + grading mesh)

Relative error 0.127%: just outside the $\pm 0.1\%$ band. M6 was 0.282%.

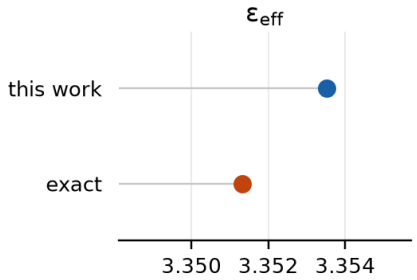
If a cleaner number is wanted

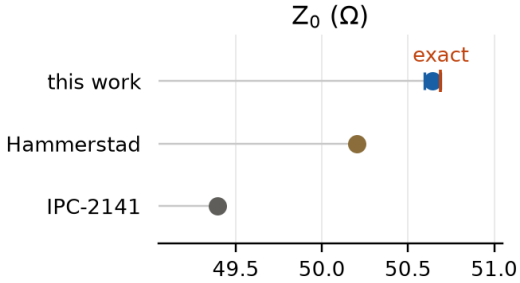
The graded mesh gives 0.017% at 139,426 unknowns; that is a different discretisation of the same model, and it is reported as such.

Takeaway

This is the number, and this is how far it can be from the truth.

ε_{eff} and Z_0 are twice as accurate as C





The ratio cancels the common bias

$\varepsilon_{\text{eff}} = C/C_{\text{air}} = 3.3535$, against an exact 3.3513: an error of 0.07%, about half the error of C alone.

Impedance against the benchmarks

$Z_0 = 1/(c_0\sqrt{C\,C_{\text{air}}}) = 50.642\,\Omega$, give or take $0.048\,\Omega$. Hammerstad gives $50.201\,\Omega$ and IPC-2141 $49.393\,\Omega$; the two formulas differ by $0.81\,\Omega$.

Takeaway

Our uncertainty ($0.048\,\Omega$) is far smaller than the spread of the formulas we compare with.

Conclusions

Established

- The solver is exact to roundoff on 23 analytic cases.
- $C_{\mathrm{ref}} = 120.467\,\mathrm{pF/m}$, agreed by two elements and two codes.
- M7 is 0.127% off, limited by the corner, at order 1.16.
- Corner grading reaches 0.017% with 39% of the unknowns.

Limits

- Truncation contributes 0.18%, comparable to the discretisation error.
- **Quasi-static model**: no losses, no radiation.
- The graded mesh is reported as an alternative, not as the submitted M7 value.

Takeaway

A numerical result is credible when its error, its source and its cure are all visible.